

UNIVERSITY OF THE ARMED FORCES



ESPE



Object Oriented Programming

SOFTWARE ENGINEERING

NRC: 30812

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1. Problem

In the day-to-day management of a convenience store, operational efficiency depends on the proper integration of all stakeholders. Currently, the management team led by Ms. [Name] faces organizational challenges, as the company's data is not digitally linked to its sales processes. The main problem is the lack of a system to manage the profile of every person who interacts with the business; this includes both customer service and monitoring the activities of the cashier on duty.

A critical issue is the disorganization of the inventory; although the products are known, there is no classification by category, which hinders sales reporting, nor is there a record of each supplier for restocking. This disorganization complicates the payment process, which requires issuing an accurate invoice that automatically deducts stock for each product. Finally, the system must ensure that each transaction concludes with a validated payment, whether by cash (calculating the exact change) or bank transfer (recording the receipt), ensuring that cash flow is transparent and free of human error.

2. Overview

In the context of managing a convenience store, daily sales operations generate critical information that must be comprehensively organized and monitored. The proposed system not only aims to automate payment processing but also to centralize management through a model that links the company with its suppliers and organizes the product catalog by category for precise control. Customers expect fast and reliable service, while the business owner requires detailed records that reflect both the cashier's performance and the actual turnover of each product. By integrating information on each individual (employees and customers) with an invoice that automatically validates inventory and processes different payment methods, such as cash and bank transfers, the risk of errors in manual calculations is eliminated. Therefore, the system offers a structural solution that improves administrative accuracy, streamlines customer service, and strengthens control over the business's revenue and inventory.

3. Background

In the context of a convenience store, the accuracy of the cash register closing is the primary indicator of good management. Historically, the use of manual calculations for cash handling and the lack of records for bank transfers have led to financial discrepancies. An automated system should allow the cashier to process sales where product information and categories are always up to date.

The system's operational foundation centers on the transition from sale to collection. When finalizing a receipt, the system must distinguish the nature of the payment: for cash, the priority is the accuracy of the change given to the customer; for bank transfers, the focus is on the traceability of the transaction number. This workflow ensures that every transaction

has a digital record, allowing the business owner to audit daily sales without having to review paper notes or informal records.

4. Analyst Comparison

The system represents real-world concepts using objects such as Customer, Product, and Invoice, which make it easier to understand and maintain; furthermore, it strategically implements inheritance by using a base class called Person, from which both the Customer and Cashier classes are derived to share common attributes without duplicating code. At the same time, the program employs polymorphism in payment processing, allowing the same method to handle a cash payment, where change is calculated, and a bank transfer, where the bank receipt is recorded, in different ways. Encapsulation is applied by protecting inventory within the Product class and delegating business logic to specific methods that prevent out-of-stock sales or incorrect total calculations. Likewise, the system effectively uses association and composition, organizing products by category and linking them to their respective suppliers, while the invoice acts as the core that integrates staff, items, and the customer into a single transaction. Finally, this structure facilitates data validation at every stage and improves the minimarket organization by transforming manual processes into an orderly, efficient, and scalable digital system.